12/08/2026

Code-E

Aakash
Medical | IIT-JEE | Foundations

Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot No. 13, Sector-18, Udyog Vihar, Gurugram, Haryana - 122015, *Ph.*+91-1244168300

MM : 720 **AIATS Prime for NEET-2027(XII Passed)_Unit Test-02** **Time : 180 Min.**

Topics Covered:
Physics : Motion in a Plane, Laws of Motion
Chemistry : Classification of Elements and Periodicity in Properties, Chemical Bonding and Molecular Structure
Botany : Biological Classification, The Living World
Zoology : Breathing and Exchange of Gases, Body Fluids and Circulation

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

PHYSICS

1. A swimmer can swim with speed of 8 m/s in still water. 800 m wide river is flowing with speed of 4 m/s. Swimmer wants to cross the river in minimum time. Velocity of swimmer with respect to ground is (approximately)

(1) 9 m/s

(2) 10 m/s

(3) 12 m/s

(4) 5 m/s

2. The angle that the vector $\overrightarrow{OA} = 5\hat{i} + 5\hat{j}$ makes with *y*-axis is

(1) 30°

(2) 45°

(3) 60°

(4) $\tan^{-1}\left(\frac{1}{2}\right)$

3. **Assertion:** A vector $(\hat{i} + \hat{j})$ is a unit vector.

Reason : $|\hat{i} + \hat{j}| = 1$

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

4. A vector $\vec{A} = \hat{i} + \sqrt{3}\hat{j}$ is rotated clockwise through an angle 15° about origin in *x*-*y* plane. The unit vector along the new vector is

(1) $\sqrt{2}\hat{i} + \sqrt{2}\hat{j}$

(2) $\frac{\hat{i}+\hat{j}}{2}$

(3) $\frac{\hat{i}}{\sqrt{2}} + \frac{\hat{j}}{\sqrt{2}}$

(4) $\frac{1}{2}\hat{i} - \frac{1}{2}\hat{j}$

5. Sum of two vectors of equal magnitudes is $\sqrt{3}$ times the magnitude of either vector. What is the angle made by the resultant vector with either vector?

(1) 60°

(2) 30°

(3) 45°

(4) 15°

6. A body moves in (*x* – *y*) plane, so that the displacement along the *x* and *y* axis are given by $x = 2t^3$ m and $y = 4t^2$ m. The speed of body at $t = 1$ s is

(1) 14 m/s

(2) 10 m/s

(3) 24 m/s

(4) 12 m/s

Scanned with OKEN Scanner

7. A particle has initial velocity $\left(4\hat{i}+6\hat{j}\right)$ m/s and a constant acceleration $\vec{a}=\left(\hat{i}+\hat{j}\right)$ m/s^2. The velocity of the particle after 2 seconds will be

(1) $\left(6\hat{i}+8\hat{j}\right)$ m/s

(2) $\left(2\hat{i}+4\hat{j}\right)$ m/s

(3) $\left(3\hat{i}+9\hat{j}\right)$ m/s

(4) $\left(3\hat{i}+\hat{j}\right)$ m/s

8. For a given speed, projectile has the same range R for two angles of projection. If t_1 and t_2 are time of flight in the two cases, then

(1) $t_1t_2 \propto R^2$

(2) $t_1t_2 \propto R$

(3) $t_1t_2 \propto \frac{1}{R}$

(4) $t_1t_2 \propto \frac{1}{R^2}$

9. Velocity of A and B are $\vec{v}_A=\left(2\hat{i}+3\hat{j}\right)$ m/s and $\vec{v}_B=\left(\hat{i}+2\hat{j}\right)$ m/s. Velocity of A as observed by B is

(1) $\left(\hat{i}+2\hat{j}\right)$ m/s

(2) $\left(\hat{i}+\hat{j}\right)$ m/s

(3) $\left(-\hat{i}-\hat{j}\right)$ m/s

(4) $\left(\hat{i}+4\hat{j}\right)$ m/s

10. Let ω_1 and ω_2 be the angular speed of the second hand and minute hand of a smoothly running analog clock respectively. If x_1 and x_2 are the respective angular displacement in 1 minute then

(1) $\frac{\omega_1}{x_1}=\frac{\omega_2}{x_2}$

(2) $\omega_1x_1=\omega_2x_2$

(3) $\omega_1x_1^2=\omega_2x_2^2$

(4) $\omega_1^2x_1=\omega_2^2x_2$

11. Choose the correct equation of trajectory for a ground to ground projectile. (Where symbols have their usual meanings)

(1) $y=x\tan\theta+\frac{gx^2}{2u^2\cos^2\theta}$

(2) $y=x\tan\theta-\frac{gx^2}{2u^2\cos^2\theta}$

(3) $y=x\tan\theta-\frac{gx}{2u^2\cos^2\theta}$

(4) $y=x\tan\theta+\frac{gx^2}{2u\cos^2\theta}$

12. A ball is thrown obliquely from the top of a tower. Then

(1) Its kinetic energy remains constant

(2) Its horizontal component of velocity remains constant

(3) Its acceleration remains constant

(4) Both (2) and (3)

13. A ball is projected with a velocity, 10 ms^{-1}, at an angle of 60° with the vertical direction. Its speed at the highest point of its trajectory will be

(1) 5 ms^{-1}

(2) 10 ms^{-1}

(3) Zero

(4) $5\sqrt{3}$ ms^{-1}

14. The angular speed of a flywheel is increased from 600 rpm to 1200 rpm in 10 s. The number of revolutions completed by the flywheel during this time is:

(1) 900

(2) 600

(3) 150

(4) 300

15. For an object thrown at 45° to the horizontal, the maximum height (H) and horizontal range (R) are related as

(1) $R=4H$

(2) $4R=H$

(3) $R=H$

(4) $R=\sqrt{2}H$

16. The velocity of a projectile projected from ground at initial point is $\left(10\hat{i}+20\hat{j}\right)$ m/s. The range of projectile is (g = 10 m/s^2)

(1) 20 m

(2) 30 m

(3) 40 m

(4) 50 m

17. The equation of motion of a projectile is $y=12x-\frac{3}{4}x^2$. What is the range of projectile? (y and x are in meter)

(1) 12 m

(2) 8 m

(3) 48 m

(4) 16 m

18. In a circular motion, the angle between the acceleration and velocity may be

(1) 90°

(2) Acute

(3) Obtuse

(4) All of these

Venom Haunt
Aakash Medical|IIT-JEE|Foundations
Scanned with OKEN Scanner

19. A projectile is projected from ground such that the angle of projection with horizontal and initial horizontal velocity are $\tan^{-1}(2)$ and 5 m/s respectively. If List-I contains time of flight (T), maximum height attained (H), horizontal range (R) and List-II has its corresponding values in appropriate SI units, then match **List-I** with **List-II**.

	List-I		List-II
a.	T	(i)	10
b.	R	(ii)	2
c.	H	(iii)	5

Choose the correct answer from the options given below:

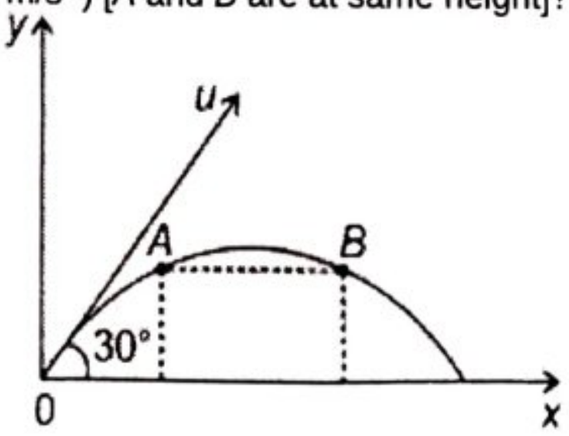
(1) a(ii), b(i), c(iii)

(2) a(i), b(ii), c(iii)

(3) a(iii), b(ii), c(i)

(4) a(i), b(iii), c(ii)

20. A particle is thrown with a velocity of u m/s. It passes through points A and B as shown in the figure at time $t_1 = 3$ s and $t_2 = 5$ s respectively. The value of u is (g = 10 m/s^2) [A and B are at same height]?



(1) 10 m/s

(2) 20 m/s

(3) 30 m/s

(4) 80 m/s

21. A boy stands on the edge of a roof 4.9 m above the ground and throws a stone horizontally with a speed of 9.8 m s^{-1}. The stone will hit the ground at an angle of θ with horizontal, where θ is

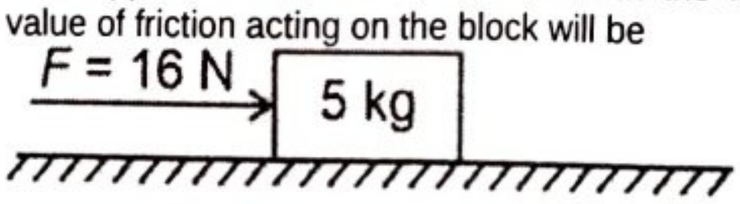
(1) 45°

(2) $\tan^{-1}(2)$

(3) $\tan^{-1}\left(\frac{1}{2}\right)$

(4) 60°

22. A block of mass m = 5 kg is placed on a rough horizontal surface having friction coefficient $\mu_S = 0.4$. If a force F = 16 N is applied on the block as shown in the figure, then value of friction acting on the block will be



(1) 20 N

(2) 16 N

(3) 8 N

(4) Zero

23. The path of a projectile fired from ground is represented by $y = \left(x - \frac{x^2}{4}\right)$ where x and y are in meter. The initial speed of the projected particle is

(1) $2\sqrt{10}$ m/s

(2) $\sqrt{10}$ m/s

(3) 2 m/s

(4) 4 m/s

24. A force applied on a mass is represented as $\vec{F} = \left(6\hat{i} - 8\hat{j} + 10\hat{k}\right)$ N and accelerates with 1 m/s^2. What will be the mass of the body?

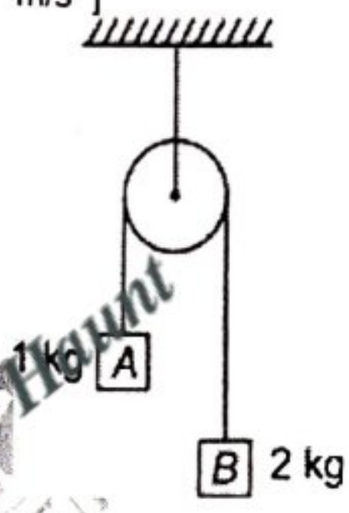
(1) 10 kg

(2) 20 kg

(3) $2\sqrt{10}$ kg

(4) $10\sqrt{2}$ kg

25. The magnitude of acceleration of the blocks A and B, connected through a light string passing over a smooth pulley, as shown in the figure, respectively are [g = 10 m/s^2]



(1) 10 m/s^2, $\frac{10}{3}$ m/s^2

(2) $\frac{10}{3}$ m/s^2, $\frac{10}{3}$ m/s^2

(3) 30 m/s^2, 10 m/s^2

(4) 4 m/s^2, 4 m/s^2

26. A machine gun fires n bullets per second with speed u, mass of each bullet is m. If bullets hit the wall and get embedded in it, then force acting on the wall is

(1) mnu

(2) $2mnu$

(3) $\frac{mnu}{2}$

(4) $4mnu$

27. A bullet is fired with a speed 200 m/s. If the mass of bullet is 100 gram and of gun is 2 kg, then the speed of gun is

(1) 20 m/s

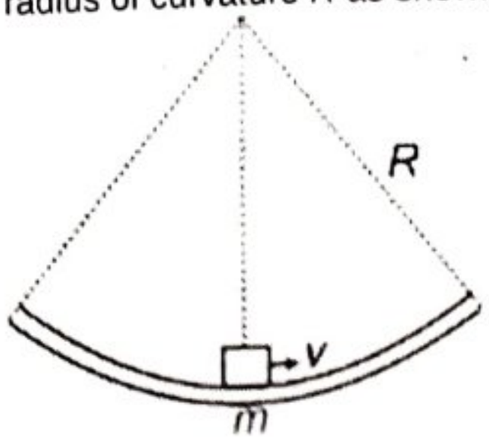
(2) 2 m/s

(3) 10 m/s

(4) 5 m/s

Scanned with OKEN Scanner

28. Normal reaction of a block on a vertical curved track of radius of curvature R as shown in figure is



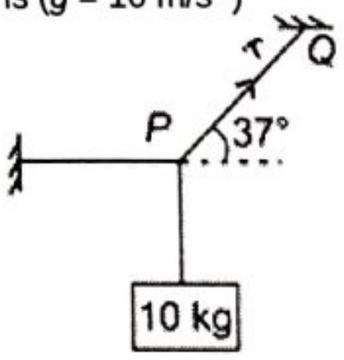
(1) $N = mg - \frac{mv^2}{R}$

(2) $N = mg + \frac{mv^2}{R}$

(3) $N = \frac{mv^2}{R} - mg$

(4) Zero

29. At equilibrium, tension in string PQ as shown in the figure is ($g = 10\ m/s^2$)



(1) 500 N

(2) $\frac{200}{3}$ N

(3) 100 N

(4) $\frac{500}{3}$ N

30. Consider the given statements and choose the correct option.

Statement A: The notion of impulse is useful when a large force acts for a short time to produce a measurable change in momentum.

Statement B: Static friction is a self adjusting force.

Statement C: Acceleration of a body at an instant does not depend on the direction of motion.

(1) Only statement A is true

(2) Only statements A and B are true

(3) Only statement C is true

(4) All statements are true

31. A block is kept on the floor of an elevator at rest. The elevator starts descending with an acceleration of $15\ m/s^2$. Speed of block at the end of one second after the start of elevator with respect to ground is (Take $g = 10\ m/s^2$)

(1) 15 m/s

(2) 30 m/s

(3) 10 m/s

(4) 5 m/s

32. If net force (in N) on a particle depends on position of particle as $F = x^2 - 6x + 8$, (where x is in m), then the distance between two points where particle will be in equilibrium, is

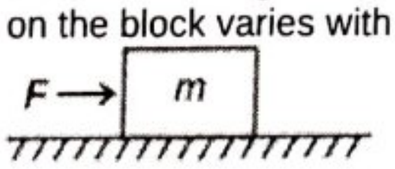
(1) 2 m

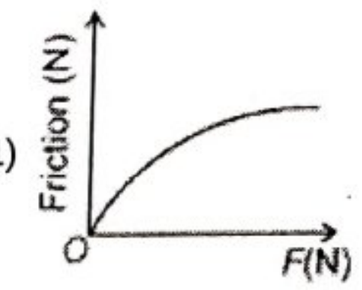
(2) 4 m

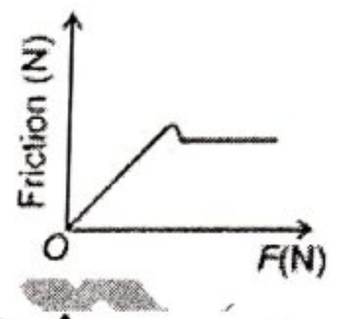
(3) 8 m

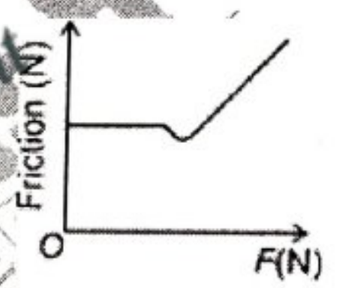
(4) 6 m

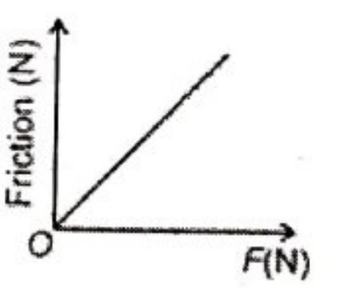
33. A block of mass m is placed on a rough horizontal surface. A variable horizontal force F is applied on the block as shown in the figure. The variation of frictional force acting on the block varies with F as



(1)


(2)


(3)


(4)


34. A car is driven round a curved path of radius 32 m without the danger of skidding. The coefficient of friction between the tyres of the car and the surface of curved path is 0.2. The maximum safe speed of the car will be

(1) 4 m/s

(2) 6 m/s

(3) 8 m/s

(4) 3 m/s

35. A metallic chain 1 m long lies on a horizontal surface of a table. The chain starts sliding on the table if 25 cm or more of it hangs over the edge of a table. The correct value of the coefficient of friction between the table and the chain is

(1) 1/3

(2) 2/3

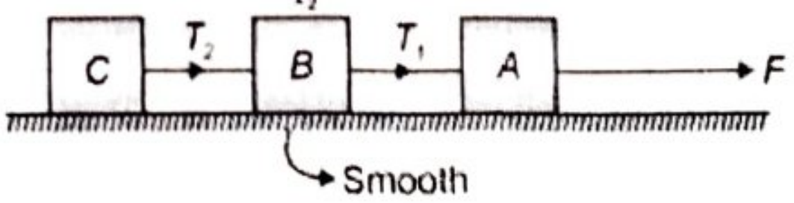
(3) 1/4

(4) 1/5

Venom Haunt Medical/IIT-JEE

Scanned with OKEN Scanner

36. Three blocks A, B and C having masses $2m$, m and $3m$ respectively, connected with ideal light strings. If a constant force F is applied on block A as shown in figure, then the value of $\frac{T_1}{T_2}$ will be



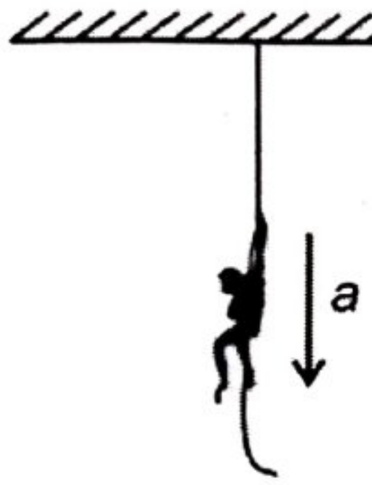
(1) $\frac{3}{2}$

(2) $\frac{4}{3}$

(3) $\frac{1}{2}$

(4) $\frac{1}{5}$

37. A monkey of mass m slides down on a string with acceleration a as shown in figure. If breaking strength of string is $\frac{2mg}{3}$, then minimum acceleration monkey can have is



(1) $\frac{g}{2}$

(2) $\frac{2g}{3}$

(3) $\frac{g}{4}$

(4) $\frac{g}{3}$

38. A person is standing on a weighing machine and observes that the reading is 60 kg, then suddenly jumps upward and observes that reading goes to 70 kg. Then his maximum upward acceleration is

(1) $\frac{g}{6}$

(2) $\frac{g}{4}$

(3) $\frac{g}{3}$

(4) $\frac{g}{2}$

39. A horizontal force 10 N is necessary to just hold a block stationary against a wall. The coefficient of friction between the block and the wall is 0.2, the weight of the block is

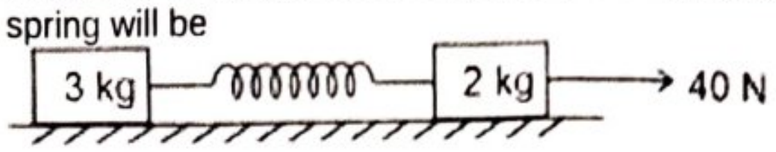
(1) 20 N

(2) 50 N

(3) 100 N

(4) 2 N

40. Two blocks of mass 3 kg and 2 kg are connected by a spring having spring constant 400 N/m. The blocks are placed on smooth horizontal surface as shown in figure. If a force of 40 N applied on 2 kg block, then extension in the spring will be



(1) 4 cm

(2) 2 cm

(3) 8 cm

(4) 6 cm

41. A particle moving on a circular path experiences centripetal force F. If the radius of circle is reduced to half and its angular speed becomes twice, then centripetal force on it becomes

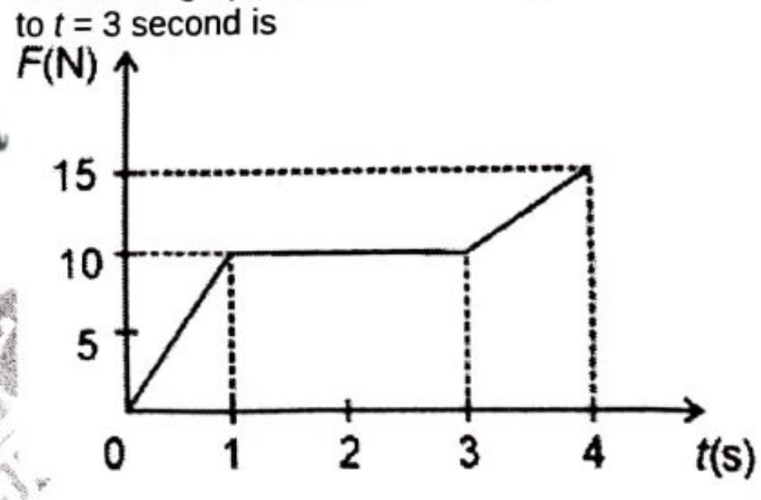
(1) $2F$

(2) $\frac{F}{2}$

(3) $8F$

(4) $4F$

42. The change in the momentum of a particle, for which force-time graph is shown in the figure, between time $t = 0$ to $t = 3$ second is



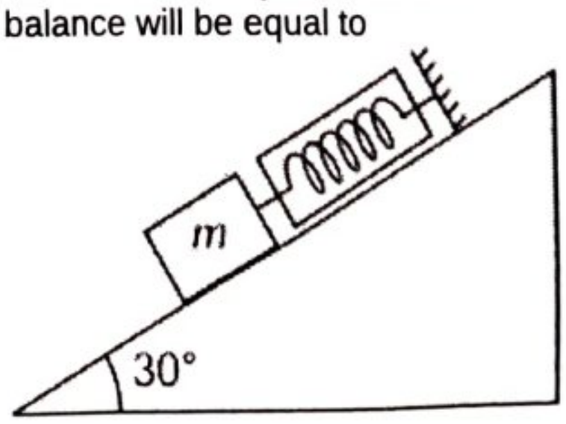
(1) 25 Ns

(2) 20 Ns

(3) 37.5 Ns

(4) 12.5 Ns

43. A body of mass 10 kg is suspended by a spring balance on an inclined plane as shown. The reading of spring balance will be equal to



(1) 25 N

(2) $50\sqrt{3}$ N

(3) 50 N

(4) 30 N

Venom Haunt
Aakash
Medical/IIT-JEE

Scanned with OKEN Scanner

44. A cyclist riding at speed of 10 m/s takes a turn on a horizontal circular road of radius $10\sqrt{3}$ m without skidding. What is his angle of inclination to the vertical?

(1) 15°

(2) 30°

(3) 45°

(4) 60°

45. The correct statement describing the action-reaction pair of forces under Newton's third law of motion is

• (1) They are equal in magnitude

(2) They may have same direction

(3) They may not act on different bodies

(4) All of these

CHEMISTRY

46. Newlands' law of octave is applicable upto

(1) Mg

(2) Al

(3) K

(4) Ca

47. The pair of elements which have same value of electronegativity on Pauling scale, is

(1) Be and B

(2) C and S

(3) Al and P

(4) Na and Mg

48. The oxidation state and covalency of Al in $[AlCl(H_2O)_5]^{2+}$ respectively are

(1) +2 and 5

(2) +3 and 3

(3) +3 and 5

• (4) +3 and 6

49. The electronic configuration of an element is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^1$. What is the atomic number of the element, which is just above the given element in the periodic table?

(1) 23

(2) 12

(3) 13

(4) 22

50. Which of the following is not a transition element?

(1) Ti

(2) V

(3) Te

(4) Fe

51. Among the following, representative element is

(1) Ni

(2) Fe

(3) Na

(4) Co

52. The correct order of first ionization enthalpy for the given elements is

(1) Be < B < C < N < O < F

(2) B < Be < C < O < N < F

(3) Be < B < C < O < N < F

(4) B < Be < C < N < O < F

53. The element which is in diagonal relationship with Be is

(1) Mg

(2) Na

(3) Al

(4) B

54. Which one of the following is correct order of the size of iodine species?

(1) $I^+ > I^- > I$

• (2) $I^- > I > I^+$

(3) $I > I^- > I^+$

(4) $I > I^+ > I^-$

55. Eka-aluminium is

(1) B

(2) Ga

(3) Si

(4) Ge

56. The symbol of element with atomic number 105 is

(1) Uup

• (2) Unp

(3) Uuu

(4) Unn

57. The element with highest negative electron gain enthalpy will belong to

(1) Group 17, period 3

(2) Group 17, period 2

(3) Group 18, period 2

(4) Group 16, period 2

Venom Hatnt
Aakash Medical/IIT-JEE|Foundations

Scanned with OKEN Scanner

58. Identify the correct order of electronegativity for the given species: (on Pauling scale)

(1) $Si < C < N < O$

(2) $Si < N < C < O$

(3) $C < Si < O < N$

(4) $C < N < Si < O$

59. Match the List-I with List-II.

List I	List II
(a) Nickel	(i) Non-metal
(b) Chlorine	(ii) Lanthanoid
(c) Germanium	(iii) Transition metal
(d) Samarium	(iv) Metalloid

Choose the correct match.

(1) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)

(2) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)

(3) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

(4) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

60. Consider the following statements:
(i) Bromine is more electronegative as compared to sulphur (on Pauling scale).
(ii) Both Barium and Boron are main group elements.
(iii) Tin and lead are metals of *p*-block.
(iv) The number of unpaired electrons in 3*d* subshell of iron is 6.
The incorrect statement(s) is/are

(1) (i) and (iv) only

(2) (iv) only

(3) (ii) only

(4) (i), (ii), (iii) and (iv)

61. Match the IUPAC names of the elements given in column I with their IUPAC symbols given in column II.

Column I	Column II
a. Unnilpentium	(i) Hs
b. Unnilhexium	(ii) Bh
c. Unnilseptium	(iii) Sg
d. Unniloctium	(iv) Db

Identify the correct match.

(1) a(iv), b(iii), c(ii), d(i)

(2) a(iv), b(i), c(ii), d(iii)

(3) a(ii), b(i), c(iv), d(iii)

(4) a(ii), b(iii), c(iv), d(i)

62. Which of the following represents the correct order of negative electron gain enthalpy for the elements O, S, Se and Te?

(1) Te < Se < S < O

(2) O < S < Se < Te

(3) O < S < Te < Se

(4) O < Te < Se < S

63. Which of the following is/are not Dobereiner's triad?
(a) Li, Na, K
(b) Cu, Ag, Zn
(c) Fe, Co, Ni
(d) Ca, Sr, Ba
Correct answer is

(1) (a) & (b) only

(2) (a) & (d) only

(3) (b) & (c) only

(4) (a), (b), (c) & (d)

64. The correct increasing order of metallic character is

(1) P < Si < Na < Mg < Be

(2) Na < Mg < Be < Si < P

(3) Be < Mg < Na < Si < P

(4) P < Si < Be < Mg < Na

65. Maximum C – O bond length is in

(1) CO

(2) CO_3^{2-}

(3) CO_2

(4) C_3O_2

66. An example of odd electron molecule is

(1) SO_3

(2) CO_2

(3) NO_2

(4) SO_2

67. Which of the following are isoelectronic and isostructural?

(1) SO_3, ClO_3^-

(2) ClO_3^-, CO_3^{2-}

(3) NO_3^-, CO_3^{2-}

(4) CO_3^{2-}, XeO_3

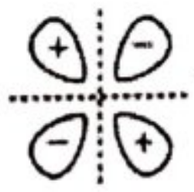
68. Which of the following is diamagnetic?

(1) NO

(2) N_2

(3) O_2

(4) N_2^+

69. 

This molecular orbital is representing:

(1) $\sigma 2p_z$ orbital

(2) $\pi^* 2p_x$ orbital

(3) $\sigma^* 2p_z$ orbital

(4) $\pi 2p_x$ orbital

Scanned with OKEN Scanner

70. Given below are two statements

Statement I : There are six hybrid orbitals in sp^3d^2 hybridisation.

Statement II : Number of hybrid orbitals are half of the number of atomic orbitals involved in hybridisation.

In light of above statements mark the correct statements

(1) Both the statement I and II are correct

(2) Both the statement I and II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

71. Given below are two statements. One is labelled as **Assertion (A)** and other is labelled as **Reason (R)**

Assertion (A): An atomic orbital is monocentric while a molecular orbital is polycentric.

Reason (R): An electron in an atomic orbital is influenced by one nucleus, in a molecular orbital it is influenced by two or more nuclei depending upon the number of atoms in a molecule.

In the light of the above statements, choose the correct answer from the options given below

•(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

72. The hybridization of atomic orbitals of xenon in XeO_3, XeF_4 and XeF_2 respectively are

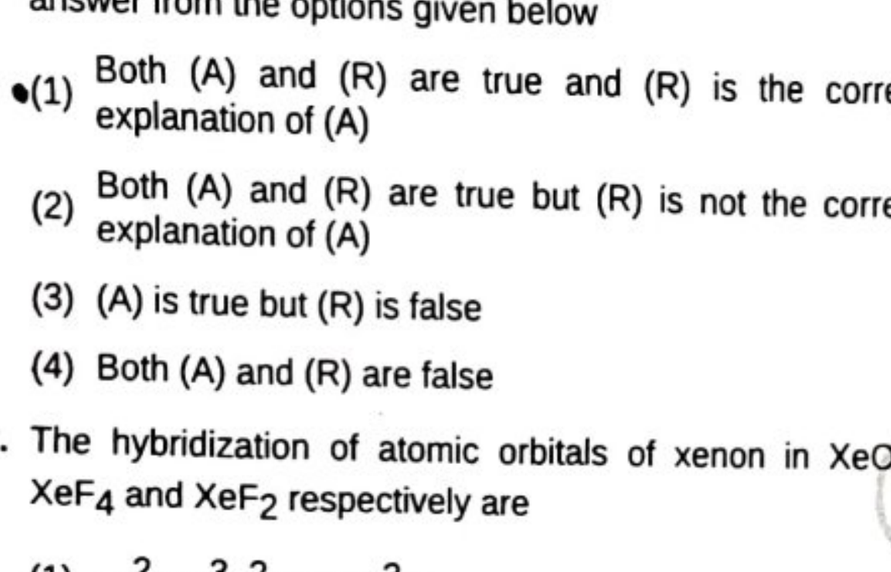
(1) sp^2, sp^3d^2 and sp^3d

(2) sp^3, sp^3d and sp^3d^2

(3) sp^3, sp^3 and sp

(4) sp^3, sp^3d^2 and sp^3d

73. The correct order of dipole moment of the following compounds is



(1) (iii) > (ii) > (i) > (iv) > (v)

(2) (ii) > (i) = (iii) > (iv) = (v)

•(3) (iii) > (i) > (ii) = (v) > (iv)

(4) (iii) > (i) > (iv) > (ii) > (v)

74. Consider the following statements:

(a) Methyl alcohol is soluble in water.

(b) Boiling point of HCl is higher than HF.

(c) Magnitude of H-bonding is maximum in solid state and minimum in the gaseous state.

(d) The hydrogen bonds have strong influence on the structure and properties of the compounds.

The correct statements are

(1) (a) and (b) only

(2) (a), (b) and (c) only

(3) (a), (c) and (d) only

(4) (a), (b), (c) and (d)

75. Shape of $N(SiH_3)_3$ and $N(CH_3)_3$ will be respectively

(1) Planar, pyramidal

(2) Pyramidal, planar

(3) Planar, planar

(4) Pyramidal, pyramidal

76. The pair of species having same bond order is

(1) N_2 and NO^+

(2) F_2 and O_2^-

(3) H_2 and He_2^+

(4) B_2 and C_2

77. Decreasing order of O–O bond length in the given species is

(1) $O_2^+ > O_2^- > O_2$

(2) $O_2 > O_2^+ > O_2^-$

(3) $O_2^+ > O_2 > O_2^-$

(4) $O_2^- > O_2 > O_2^+$

78. The correct order of bond length of C – C, C – N, C – H is

(1) C – H < C – N < C – C

(2) C – C < C – N < C – H

(3) C – C < C – H < C – N

•(4) C – N < C – H < C – C

79. Consider the following statements

(a) Shape of ozone molecule is bent.

(b) Ozone has 3 canonical forms.

(c) Formal charge on central oxygen atom in ozone is zero.

Choose the incorrect statement.

(1) (a) and (b) only

•(2) (c) only

(3) (b) only

(4) (b) and (c) only

Scanned with OKEN Scanner

80. All of the given species have equal number of σ and π bonds, except

(1) CO_2

(2) SO_2

(3) SO_3

(4) C_2H_2

81. Shape and electronic geometry around central atom of I_3^- ion are respectively

(1) Linear and trigonal planar

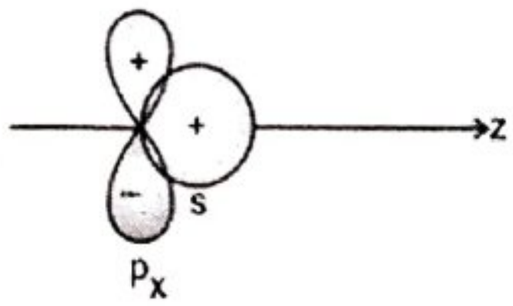
(2) Trigonal bipyramidal and trigonal bipyramidal

(3) Linear and trigonal bipyramidal

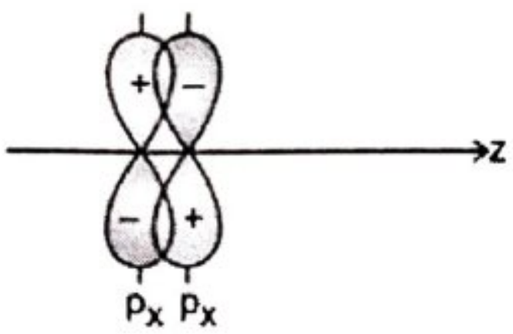
(4) Bent and distorted tetrahedral

82. Which of the given overlapping of orbitals leads to zero overlapping?

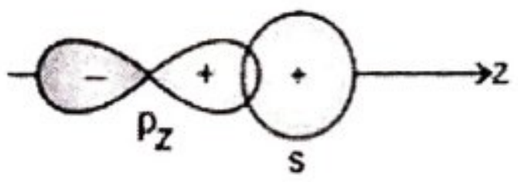
(1)



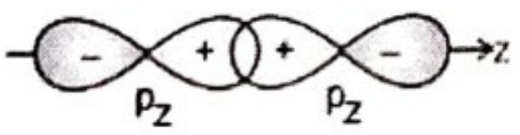
(2)



(3)



(4)



83. Consider the following statements:
(i) NaCl is more covalent than $MgCl_2$
(ii) CsI is more ionic as compared to NaI
(iii) Al^{3+} has more polarising power than Mg^{2+}
The **incorrect** statement(s) is/are

(1) (i) only

(2) (ii) only

(3) (ii) & (iii) only

(4) (i) & (iii) only

84. The correct order of repulsive interaction of electron pairs is
lp = lone pair
bp = bond pair

(1) lp – bp > bp – bp > lp – lp

(2) lp – lp > lp – bp > bp – bp

(3) lp – lp > bp – bp > lp – bp

(4) bp – bp > lp – bp > lp – lp

85. If the observed dipole moment of A–B molecule is 0.8 D and A–B bond length is 100 pm then percentage ionic character in A–B molecule will be nearly

(1) 25.75%

(2) 16.65%

(3) 42.25%

(4) 31.85%

86. Match List-I with List-II

	List-I (Species)		List-II (Bond angle)
(a)	BF_3	(i)	Contains only 180° and 90° bond angles
(b)	XeF_4	(ii)	Contains only 120° bond angle
(c)	PCl_5	(iii)	Contains 90°, 180° and 120° bond angles
(d)	CO_2	(iv)	Contains only 180° bond angle

The correct match is

(1) a(ii), b(iii), c(i), d(iv)

(2) a(i), b(iii), c(ii), d(iv)

(3) a(ii), b(i), c(iii), d(iv)

(4) a(i), b(ii), c(iv), d(iii)

87. Which of the following overlaps is incorrect? (assuming z-axis to be the internuclear axis)

(1) $2p_y + 2p_y \rightarrow \pi 2p_y$

(2) $2p_z + 2p_z \rightarrow \sigma 2p_z$

(3) $2p_x + 2p_x \rightarrow \pi 2p_x$

(4) $1s + 2p_y \rightarrow \pi(1s - 2p_y)$

88. Highest occupied molecular orbital (HOMO) of F_2 molecule is

(1) σ

(2) σ^*

(3) π

(4) π^*

89. Identify a molecule which does not exist.

(1) Li_2

(2) C_2

(3) O_2

(4) He_2

90. Among of the following ionic compounds, the one that has maximum lattice enthalpy is

(1) NaF

(2) NaCl

(3) LiCl

(4) LiF

Scanned with OKEN Scanner

K P C O F G S

BOTANY

91. Binomial nomenclature was given by

(1) Robert Hooke

(2) Robert Brown

●(3) Carolus Linnaeus

(4) Rudolf Virchow

92. The specific epithets, *nigrum* and *melongena* belongs to the order

(1) Poales

(2) Sapindales

●(3) Polymoniales

(4) Solanaceae

93. Observe the following taxa associated with wheat:
a. Poales
b. Poaceae
c. Angiospermae
d. Plantae
e. Monocotyledonae
Which of the following options **correctly** depicts the sequential order of these taxa in ascending order?

(1) a → d → b → e → c

●(2) b → a → e → c → d

(3) b → e → c → a → d

(4) c → a → d → b → e

94. Identify the group of 'taxa' belonging to the same taxonomic category

(1) *Triticum*, Poaceae, Poales

(2) Anacardiaceae, Sapindales, Dicotyledonae

●(3) Diptera, Sapindales, Poales

(4) Poaceae; Solanaceae, *Petunia*

95. Systematics differ from modern taxonomy, as it takes into account

(1) External structure

(2) Structure of cell

(3) Developmental process

●(4) Evolutionary relationship

96. Match the following columns and select the **correct** option.

	Column-I		**Column-II**
a.	Man	(i)	Anacardiaceae
b.	Housefly	(ii)	Muscidae
c.	Mango	(iii)	Poaceae
d.	Wheat	(iv)	Hominidae

●(1) a(iv), b(ii), c(i), d(iii)

(2) a(ii), b(iii), c(iv), d(i)

(3) a(iv), b(ii), c(iii), d(i)

(4) a(iii), b(i), c(ii), d(iv)

97. Read the following statements and select the correct option.
Statement A: Taxonomic categories are distinct biological entities and not merely morphological aggregates.
Statement B: Classification of organisms into various categories does not help in identifying similarities and dissimilarities among different organisms.

(1) Only A is incorrect

●(2) Only B is incorrect

(3) Both A and B are correct

(4) Both A and B are incorrect

98. Read the following statements and choose the option which is **true** for them
Statement A : The number of similar characteristics goes on increasing from species to kingdom.
Statement B : Higher the category, greater is the difficulty of determining the relationship to other taxa at same level.

(1) Only statement A is correct

●(2) Only statement B is correct

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

99. According to IUCN, currently the known and described species of all organisms are between

(1) 0.17-0.18 million

●(2) 1.7-1.8 million

(3) 17-18 million

(4) 170-180 million

100. The equivalent taxonomic categories of Diptera and Insecta w.r.t. plants are respectively

(1) Anacardiaceae and Sapindales

(2) Poales and Angiospermae

●(3) Sapindales and Dicotyledonae

(4) Monocotyledonae and Angiospermae

101. The **correct** binomial epithet of mango is

(1) *Mangifera indica*

●(2) *Mangifera indica* Linn

(3) *Mangifera indica Linn*

(4) *Magnifera Indica* Linn

102. Which of the given features for dinoflagellates is **incorrect**?

(1) They appear yellow, green, brown, blue or red depending on the main pigments present in their cells.

(2) Unicellular, motile and flagellated

(3) Mainly are marine but some are fresh water forms

●(4) Do not contain photosynthetic pigments

Scanned with OKEN Scanner

103. Diptera, Primata and Sapindales are

(1) Groups of related classes comprising of organisms with common feature like presence of notochord.

(2) Groups of taxa that comes in between division and class

●(3) Groups of related divisions with still less number of similarities as compared to genus and species.

(4) Assemblage of families which exhibit a few similar characters and similar characters are less in number as compared to different genera included in a family.

104. Match the following columns and select the **correct** option.

	Column I		Column II
a.	Taxonomy	(i)	These are distinct biological entities and not merely morphological aggregates
b.	Systematics	(ii)	A process of classification, based on characteristics, all living organisms can be classified into different taxa
c.	Classification	(iii)	It takes into account evolutionary relationships between organisms
d.	Taxonomic categories	(iv)	Process by which anything is grouped into convenient categories based on some easily observable characters

(1) a(iii), b(iv), c(i), d(ii)

●(2) a(ii), b(iii), c(iv), d(i)

(3) a(ii), b(i), c(iv), d(iii)

(4) a(iv), b(iii), c(i), d(ii)

105. The immediate next lower category to kingdom in taxonomic hierarchy for plants is

(1) Genus

● (2) Division

(3) Class

(4) Order

106. Mango and potato belong to same

(1) Family

●(2) Class

(3) Genera

(4) Order

107. Which of the following is **not** the characteristic of protozoa ?

(1) Pseudopodia can be found as a locomotory structure in some of the members

(2) Binary fission is used as a means of reproduction by some of the members

(3) Contractile vacuole can be found as a part of endomembrane system in some of the members

(4) Chitinous cell wall is the outer boundary in all the members

108. Blue green algae resembles with green plants due to

●(1) Presence of chlorophyll 'a'

(2) Chemosynthetic nutrition

(3) Cell wall structure

(4) Nucleus organisation

109. Match the following

	Column-I (Group of bacteria)		Column-II (Their shape)
a.	Coccus	(i)	Rod-shaped
b.	Bacillus	(ii)	Spherical
c.	Spirillum	(iii)	Spiral
d.	Vibrium	(iv)	Comma-shaped

(1) a(i), b(ii), c(iii), d(iv)

●(2) a(ii), b(i), c(iii), d(iv)

(3) a(i), b(ii), c(iv), d(iii)

(4) a(ii), b(i), c(iv), d(iii)

110. The body of *Euglena* is flexible due to the presence of

(1) Cell wall rich in polypeptides

●(2) Protein rich layer called pellicle

(3) Cilia all around the cell envelope

(4) Proteinless plasma membrane

111. Match the following columns and select the **correct** option.

Column I		Column II
(a) *Penicillium*	(i)	Belongs to a fungal class whose members are saprophytes or parasites while a large number of them are decomposers of litter
(b) *Rhizopus*	(ii)	Belongs to a fungal class in which asexual spores are generally not found but vegetative reproduction occur by fragmentation
(c) *Ustilago*	(iii)	Belongs to a fungal class in which sexual spores are produced endogenously in sac like asci
(d) *Alternaria*	(iv)	Belongs to a fungal class in which asexual spores are produced endogenously in sporangium

(1) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(2) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

●(3) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(4) (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)

112. *Cuscuta* is a/an

(a) Eukaryotic organism
(b) Heterotrophic organism
(c) Parasite
(d) Plant

(1) Only (a) is correct

(2) Only (a) & (d) are correct

(3) Only (a), (b) & (d) are correct

●(4) All (a), (b), (c) & (d) are correct

Scanned with OKEN Scanner

113. The causative agent of potato spindle tuber disease is

(1) Virus

(2) Prion

(3) Viroid

(4) Lichen

114. Which of the following matches is **not** correct?

	Feature	Phycomycetes	Ascomycetes	Basidiomycetes
(1)	Mycelium	Aseptate, coenocytic	Septate, branched	Septate, branched
(2)	Asexual spore	Zoospore	Conidia	Generally absent
(3)	Sexual spore	Zygospore	Ascospore	Absent
(4)	Fruiting bodies	Absent	Ascocarp	Basidiocarp

(1) (1)

(2) (2)

(3) (3)

(4) (4)

115. Select the **incorrect** statement about kingdom Protista.

(1) Members of Protista are primarily aquatic

(2) Boundaries of this kingdom are not well defined

(3) Flagella and cilia have (9 + 2) organisation of microtubules

(4) Organisation at tissue level is present

116. Read the following statements and select the correct option.

Statement-1 : All parasitic bacteria are heterotrophic but all heterotrophic bacteria are not parasitic.

Statement-2 : All N_2-fixing bacteria are symbiotic but all symbiotic bacteria are not N_2-fixing.

(1) Only statement-1 is correct

(2) Only statement-2 is correct

(3) Both the statements are correct

(4) Both the statements are incorrect

117. State true **(T)** or false **(F)** for the following statements and select the correct option.

(a) Fungi prefer to grow in cold and moist places

(b) Most fungi absorb soluble organic matter from dead substances.

(c) Members of phycomycetes could be found in aquatic habitats.

	(a)	(b)	(c)
(1)	T	T	F
(2)	T	F	T
(3)	F	T	T
(4)	F	T	F

(1) (1)

(2) (2)

(3) (3)

(4) (4)

118. Select the **correct** option regarding the sequential steps occur during sexual cycle in some members of Phycomycetes.

(1) Plasmogamy → Karyogamy → Meiosis in zygospore

(2) Karyogamy → Plasmogamy → Meiosis in zygospore

(3) Dikaryophase → Plasmogamy → Karyogamy

(4) Fusion of nuclei → Fusion of protoplasm → Dikaryophase

119. Match the following columns and select the **correct** option.

	Column I		Column II
a.	*Ustilago*	(i)	Zygospores
b.	*Claviceps*	(ii)	Basidiospores
c.	*Mucor*	(iii)	Conidia
d.	*Trichoderma*	(iv)	Ascospores

(1) a(ii), b(i), c(iv), d(iii)

(2) a(iii), b(iv), c(i), d(ii)

(3) a(ii), b(iii), c(iv), d(i)

(4) a(ii), b(iv), c(i), d(iii)

120. Match column A with column B and select the **correct** option.

	Column A		Column B
a.	Morels	(i)	Aseptate and coenocytic mycelium
b.	*Puccinia*	(ii)	Edible fruiting body
c.	*Alternaria*	(iii)	Rust fungus
d.	*Albugo*	(iv)	Only vegetative phase is known

(1) a(ii), b(iii), c(iv), d(i)

(2) a(i), b(ii), c(iii), d(iv)

(3) a(ii), b(i), c(iv), d(iii)

(4) a(iv), b(iii), c(ii), d(i)

121. Viroids differ from viruses in having

(1) DNA molecules with protein coat

(2) DNA molecules without protein coat

(3) RNA molecules with protein coat

(4) RNA molecules without protein coat

122. Read the statements I and II and select the correct option.

Statement I: All the unicellular organisms are placed in kingdom Protista.

Statement II: Kingdom Protista has brought together unicellular photosynthetic organisms like *Chlamydomonas*, *Chlorella*, *Spirogyra* with *Paramoecium* and *Amoeba*.

In the light of above statements, choose the **most appropriate** answer from the options given below:

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

Scanned with OKEN Scanner

123. Creutzfeldt-Jacob disease in humans is caused by

(1) A virus that also causes the mad cow disease in cattle.

(2) An infectious agent that is similar in size to the causative agent of citrus canker.

(3) A pathogen that was found to be a free RNA.

(4) An analogous variant of the infectious agent that causes bovine spongiform encephalopathy in cattle.

124. A zygospore is

(1) Flagellated, motile asexual spore

(2) Resting, non-motile, diploid spore

(3) Non flagellated, non-motile asexual spore

(4) Flagellated sexual spore of sac fungi

125. The three-domain system of classification of organisms

(1) Includes Protista, Fungi, Plantae and Animalia in one domain

(2) Divides Protista into two domains

(3) Includes Monera and Protista in one domain

(4) Includes all autotrophs in one domain

126. Slime moulds show similarity with plants and animals as

(1) They have cell wall composed of peptidoglycan and do not reproduce asexually

(2) They don't have cell wall at any stage of life and show heterotrophic mode of nutrition

(3) They have cellulosic cell wall in their spores and main body plasmodium is without cell wall

(4) They have cell wall composed of chitin in spores and main body lacks cell wall

127. The body of fungi consists of long, slender thread-like structures called ___(i)___ and the network of ___(ii)___ is known as ___(iii)___.
Select the **correct** option to fill in the blanks (i), (ii) and (iii).

(1) (i) & (iii) - Mycelia (ii) - Filament

(2) (i) & (iii) - Filament (ii) - Thallus

(3) (i) & (ii) - Hyphae (iii) - Mycelium

(4) (i) - Thalli (ii) & (iii) - Hypha

128. The basidiospores

(1) Are endogenously produced spores

(2) Are produced on the basidium

(3) Are asexual spores

(4) Are diploid in nature

129. Slime moulds

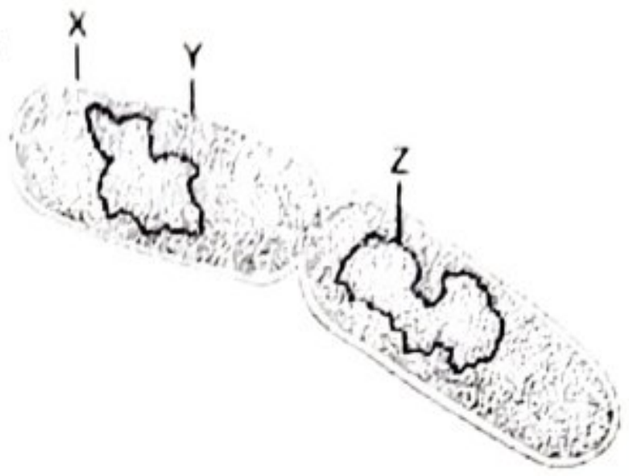
(1) Are multicellular organisms

(2) Form plasmodium under unfavourable conditions

(3) Are saprophytic fungi

(4) Produce spores which possess true walls

130.



Select the option that **correctly** depicts the above labels of the given organism.

(1) 'X' represents the cellulosic cell wall consisting of polysaccharide, along with amino acid

(2) 'Y' represents the capsule, which protects the organism from bursting and collapsing

(3) 'Z' represents the genetic material of the organism which is a circular DNA

(4) 'Y' represents the mucilaginous sheath that covers the cell wall of the organism

131. Members of deuteromycetes differ from the members of ascomycetes as the former

(1) Have aseptate mycelium

(2) Have endogenously produced asexual spores

(3) Lack sexual spores

(4) Lack parasitic members

132. Primitive bacteria called methanogens

(1) Are photoautotrophic bacteria

(2) Have peptidoglycan as major component in their cell wall

(3) Are abundant in marshy areas

(4) Show aerobic respiration

133. Vast majority of bacteria are

(1) Phototrophic

(2) Chemosynthetic

(3) Autotrophic

(4) Heterotrophic

134. Lichens are the symbiotic association between

(1) Two photosynthetic organisms

(2) A phycobiont and a mycobiont

(3) Two non-photosynthetic organisms

(4) A fungus and a higher plant

135. Viruses could be crystallised, was shown by

(1) D. J. Ivanowsky

(2) M. W. Beijerinck

(3) W. M. Stanley

(4) T. O. Diener

Scanned with OKEN Scanner

ZOOLOGY

136. The gaseous product of cellular respiration is carried by haemoglobin as

•(1) Carbamino-haemoglobin

(2) Oxyhaemoglobin

(3) Methaemoglobin

(4) Carboxyhaemoglobin

137. During inspiration,

•(1) Diaphragm contracts to increase thoracic volume

(2) Diaphragm relaxes to decrease thoracic volume

(3) No change in position of diaphragm or thoracic volume is seen

(4) Ribs and sternum remain in their original position

138. The maximum volume of air a person can breathe in after a forced expiration is about

(1) 1100 mL

(2) 8000 mL

(3) 500 mL

•(4) 4500 mL

139. Choose the **incorrect** statement regarding blood pressure.

•(1) Diastolic pressure is always greater than systolic pressure

(2) Blood pressure is greater in aorta than the inferior vena cava

(3) Normal blood pressure is registered as Systolic/Diastolic mmHg

(4) 200/100 mmHg pressure can cause damage to the vital organs

140. Read the given statements:

(A) Human heart is myogenic.

(B) AV node can generate maximum number of action potentials (70-75/minute) in human heart.

For the above statements which one of the following option is **correct**?

(1) A is incorrect but B is correct

•(2) A is correct but B is incorrect

(3) A and B, both are correct

(4) A and B, both are incorrect

141. In case of a standard electrocardiogram, the heart rate of a healthy individual can be determined by counting the number of _____, representing ventricular depolarisation, that occur in a given time period. Select the option that correctly fills the blank.

(1) P-waves

(2) T-waves

•(3) QRS complexes

(4) PQ intervals

142. Select the **incorrect** statement w.r.t. lymph.

(1) It is a colourless fluid containing lymphocytes.

(2) It is an important carrier for nutrients, hormones, *etc.*

•(3) It has different mineral distribution as that present in plasma.

(4) Fats are absorbed through it in the lacteals present in the intestinal villi.

143. A person tries to hold his breathe for sometime. Which of the following changes in blood would first lead to an urge to breathe?

(1) Fall in O_2 concentration

•(2) Rise in CO_2 concentration

(3) Rise in pH

(4) Rise in O_2 concentration

144. Select the **correct** statement w.r.t. formed elements of blood

•(1) Basophils secrete histamine, serotonin and heparin and are thus associated with inflammatory reactions

(2) Lymphocytes constitute about 2-5% of total WBCs

(3) Neutrophils are agranular phagocytic cells

(4) Both B and T lymphocytes are responsible for immune responses of the body and both undergo maturation within the bone marrow only

145. **Assertion (A) :** On activation of parasympathetic nervous system, cardiac output decreases.

Reason (R) : Parasympathetic nerve endings release adrenaline which decreases rate of heart beat and speed of conduction of action potential.

In the light of above statements, select the **correct** option.

(1) Both Assertion and Reason are false statements

(2) Assertion is true statement but Reason is false

(3) Both Assertion & Reason are true but reason is not the correct explanation of assertion

(4) Both Assertion & Reason are true and the reason is the correct explanation of assertion

146. Match column I with column II and select the **correct** option.

	Column I		Column II
a.	FRC	(i)	ERV + RV
b.	VC	(ii)	RV + ERV + IRV + TV
c	IC	(iii)	TV + IRV
d.	TLC	(iv)	ERV + TV + IRV

•(1) a(i), b(iv), c(iii), d(ii)

(2) a(i), b(ii), c(iii), d(iv)

(3) a(i), b(iv), c(ii), d(iii)

(4) a(i), b(ii), c(iv), d(iii)

Scanned with OKEN Scanner

147. Rh incompatibility occurs when mother is __A__, biological father is __B__ and foetus is __C__.
Here A, B, C respectively are

(1) Rh –ve, Rh +ve, Rh +ve

(2) Rh +ve, Rh –ve, Rh –ve

(3) Rh +ve, Rh –ve, Rh +ve

(4) Rh –ve, Rh +ve, Rh –ve

148. The function of chordae tendineae is to

(1) Close the mitral valves

(2) Close the semilunar valves

(3) Close the atrioventricular valves during joint diastole

(4) Prevent AV valves from prolapsing into atria during ventricular contraction

149. **Assertion (A) :** The cardiac output of an athlete will be more during exercise compared to that of an ordinary man.
Reason (R) : Cardiac output is based on the body's ability to alter the stroke volume as well as the heart rate.
In the light of above statements, select the **correct** option.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion.

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion.

(3) Assertion is true statement but Reason is false.

(4) Both Assertion and Reason are false statements.

150. Higher number of RBCs in the blood of a mountain dweller in comparison to a person residing in plains can be explained by

(1) Lack of pollutants in higher mountainous regions

(2) Less pO_2 in atmospheric air on mountains requiring more RBCs to supply the required amount of O_2 to tissues

(3) Better nutrition intake of mountain dwellers

(4) Enhanced erythropoiesis in mountain dwellers due to high exposure to UV rays

151. **Assertion (A):** Blood is a connective tissue containing plasma and formed elements.
Reason (R) : It is the main circulating fluid that helps in transport of various substances.
In the light of above statements select the **correct** option.

(1) Both Assertion and Reason are true and Reason is the correct explanation of the Assertion

(2) Both Assertion and Reason are true but Reason is not the correct explanation of the Assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

152. Aortic and carotid chemoreceptors detect changes in

(1) CO_2 and H^+ concentration

(2) Body temperature

(3) Partial pressure of CO_2 only

(4) O_2 concentration

153. Select the **incorrect** statement w.r.t. the nodal tissue of the heart.

(1) The entire nodal tissue, from sino-atrial node to the Purkinje fibres has the ability to generate action potentials.

(2) All parts of the nodal tissue generate the same number of action potentials per minute.

(3) Both sino-atrial and atrio-ventricular nodes are present in the right atrium.

(4) Sino-atrial node normally acts as the pacemaker.

154. Thoracic chamber is ventrally, dorsally and laterally formed respectively by

(1) Vertebral column, ribs, and sternum

(2) Ribs, sternum, and diaphragm

(3) Sternum, vertebral column, and ribs

(4) Diaphragm, sternum, and ribs

155. Structures which are **not** supported by incomplete cartilaginous rings are

(1) Tracheae

(2) Primary bronchi

(3) Initial bronchi

(4) Terminal bronchioles

156. Select the **correct** match between respiratory volume/capacity and their value w.r.t. a normal adult human.

(1) Residual volume : 500 mL

(2) Vital capacity : 1000-1100 mL

(3) Inspiratory capacity : 2300-7000 mL

(4) Functional residual capacity : 2100-2300 mL

157. _____ respiration is seen in most aquatic arthropods and molluscs.
Select the option that fills the blank **correctly**

(1) Pulmonary

(2) Tracheal

(3) Cutaneous

(4) Branchial

158. Below given are the features of 'X' involved in the regulation of respiration.
(a) Neural signals from this centre can reduce the duration of inspiration.
(b) It can moderate the functions of respiratory rhythm centre.
(c) It is present in the pons region of the brain.
Select the option that correctly identifies 'X'.

(1) Pneumotaxic centre

(2) Respiratory rhythm centre

(3) Chemosensitive area

(4) Appetite centre

Venom Hunt Aakash Medical IIT JEE Foundations

Scanned with OKEN Scanner

159. Which of the following **cannot** be measured by spirometer?
(i) FRC
(ii) RV
(iii) TLC

(1) Only (i)

•(2) (i), (ii) & (iii)

(3) Only (ii) & (iii)

(4) Only (i) & (iii)

160. Individuals with which blood group lack both A and B antigens on the surface of their RBCs?

•(1) O –ve

(2) AB +ve

(3) A –ve

(4) B +ve

161. Angina pectoris refers to

(1) Adequate blood supply to heart

(2) Deposition of plaque in pulmonary vein

(3) Increase in blood pressure

•(4) Acute chest pain

162. Every 2 L of oxygenated blood can deliver about 'X' mL of oxygen to the tissues under normal conditions in humans. Choose the option which represents **correct** value for 'X'.

(1) 80

•(2) 100

(3) 12

(4) 120

163. Choose the **incorrect** statement.

(1) Plasma is a viscous fluid constituting nearly 55% of the blood

(2) Fibrinogen is involved in clotting of the blood

•(3) Plasma with active clotting factors is called serum

(4) Formed elements constitute nearly 45 per cent of blood

164. Choose the **incorrect** statement w.r.t. blood vessels.

(1) Arteries are non collapsible as they have thick walls

•(2) Veins do not have valves in them

(3) Most of the veins carry deoxygenated blood towards the heart

(4) Through capillaries, exchange of gases and nutrients occurs between blood and tissues

165. All of the following functions are performed by conducting part of human respiratory system, **except**

(1) Transportation of air into the alveoli

(2) Clearing atmospheric air from the foreign particles

•(3) Gaseous exchange between the blood and atmospheric air

(4) Bringing atmospheric air to the body temperature

166. What is the partial pressure of oxygen and carbon dioxide in the systemic artery respectively?

	Oxygen	Carbon dioxide
(1)	40 mm Hg	45 mm Hg
(2)	95 mm Hg	40 mm Hg
(3)	104 mm Hg	40 mm Hg
(4)	45 mm Hg	95 mm Hg

(1) (1)

•(2) (2)

(3) (3)

(4) (4)

167. The state of heart when it is not pumping blood effectively enough to meet the needs of the body is called

•(1) Cardiac arrest

(2) Heart failure

(3) Heart attack

(4) Angina pectoris

168. Read the following statements w.r.t. an adult human, under normal physiological conditions.
a. Partial pressure of oxygen in deoxygenated blood is 40 mm Hg
b. Low temperature favours the formation of oxyhaemoglobin
c. Oxygen dissociation curve shifts to the right side at tissue level
d. Haemoglobin is 75% saturated with oxygen in deoxygenated blood.
Choose the option with all the **correct** statements.

(1) b and c only

(2) a and d only

(3) a, c and d only

(4) a, b, c and d

169. **Assertion (A) :** Portal system consists of portal vein which forms a network of capillaries in another or intermediate organ before reaching the heart.
Reason (R) : Renal portal system is present in all vertebrates.
In the light of above statements, choose the **correct** option.

(1) Both Assertion and Reason are false statements

•(2) Assertion is true statement but Reason is false

(3) Both Assertion & Reason are true but reason is not the correct explanation of assertion

(4) Both Assertion & Reason are true and the reason is the correct explanation of assertion

170. During coagulation of blood the conversion of inactive fibrinogen into fibrin is catalysed by

(1) Thromboplastin

(2) Proconvertin

(3) Thrombokinase

•(4) Thrombin

Scanned with OKEN Scanner

171. Read the statements given below and select the **correct** option.

(1) Interstitial fluid has the same mineral distribution as that in plasma

(2) Electrocardiograph is a graphical representation of the electrical activity of the heart during a cardiac cycle

(3) The body does not have the ability to alter the duration of cardiac cycle but it can change its stroke volume and heart rate

(4) Cardiac cells forming SA node generate action potentials at rate of 100 times/min in presence of acetylcholine in a normal healthy person

172. Steps involved in respiration are listed below. Arrange them in correct order and select the **correct** option.

a. Utilisation of O_2 by tissues

b. Breathing/inspiration

c. Exchange of gases between oxygenated blood and tissue

d. Exchange of gases between deoxygenated blood and alveoli

(1) d → c → b → a

(2) b → c → d → a

(3) a → b → c → d

(4) b → d → c → a

173. The rate of breathing in a healthy adult human under normal physiological conditions is

(1) 38-44 times/min

(2) 12-16 times/min

(3) 70-72 times/min

(4) 20-24 times/min

174. Respiration is chiefly controlled by

(1) Cerebellum

(2) Medulla oblongata

(3) Olfactory lobes

(4) Hypothalamus

175. Given are the events in a cardiac cycle. Mark the **incorrect** option.

(1) During ventricular relaxation, the opening of cuspid valves precedes the closure of semilunar valves

(2) Semilunar valves are forced open during ventricular systole

(3) Action potential from SAN stimulates both atria to undergo a simultaneous contraction

(4) Bundle of His transmits stimuli received from AVN through ventricular musculature

176. Read the following statements A and B and choose the **correct** option w.r.t. humans.

Statement A: Movement of air into and out of the lungs is due to a pressure gradient between the lungs and the atmosphere.

Statement B: Inspiration takes place when there is a positive pressure inside the lungs with respect to atmospheric air.

(1) Statement A is correct and B is the correct explanation of A

(2) Statement A is correct but B is incorrect

(3) Both statements A and B are correct

(4) Statement A is incorrect but B is correct

177. Choose the **incorrect** statement.

(1) Heart sound 'dub' is high pitched and of shorter duration than 'lub'.

(2) Heart sound 'lub' is produced by closing of semilunar valves at the beginning of ventricular diastole.

(3) The volume of blood pumped out by each ventricle per minute is known as the cardiac output.

(4) Cardiac cycle includes systole and diastole of both the atria and ventricles.

178. Read the following statements w.r.t. respiratory system in humans.

(A) At the tissue site, carbonic acid dissociates into H_2O and CO_2 in presence of enzyme carbonic anhydrase.

(B) Diffusion membrane is made up of three major cellular layers.

Choose the **correct** option.

(1) Both statements A and B are correct

(2) Only statement A is correct

(3) Only statement B is correct

(4) Both statements A and B are incorrect

179. During joint diastole, all the four chambers of the heart are in

(1) Contracted state

(2) Relaxed state

(3) Partially contracted state

(4) Empty state

180. The important function of valves in the heart is to

(1) Encourage backward flow of blood

(2) Change the direction of blood flow

(3) Prevent the backward flow of blood

(4) Pump the blood strongly to farther regions of the body

Venom Haunt Aakash Medical IIT-JEE Foundations

Scanned with OKEN Scanner